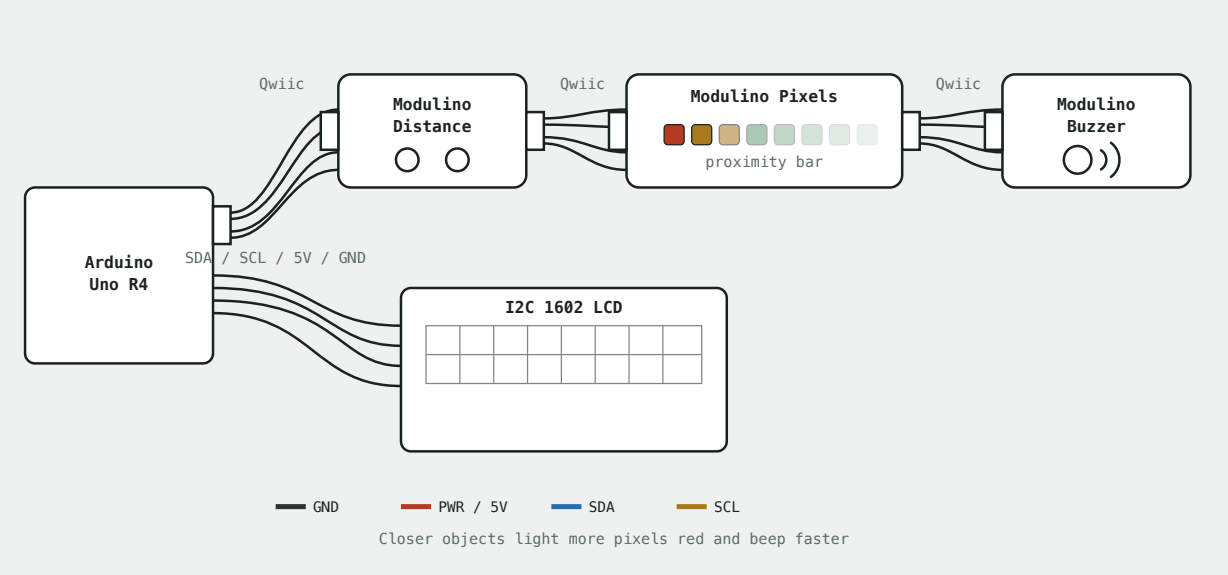
Distance-sensing parking assistant

Four modules at once, acting like a car's parking sensor: Modulino Distance measures range, Pixels lights a proximity bar, the Buzzer beeps faster as things get closer, and the LCD shows the live number.

Board · Arduino Uno R4 **Modules** · Distance, Pixels, Buzzer, I2C LCD
New concepts · `map()` / `constrain()`, non-blocking timing



Three Modulino modules chain in a row (order doesn't matter); the LCD is wired exactly as in **project 2**. Closer objects light more pixels toward red and shorten the gap between beeps.

PIN & CONNECTION REFERENCE

Component	Connects to
Modulino Distance	Qwiic/I2C bus
Modulino Pixels	Qwiic/I2C bus
Modulino Buzzer	Qwiic/I2C bus
I2C 1602 LCD	VCC→5V, GND→GND, SDA→SDA, SCL→SCL (same as project 2)

REQUIRED LIBRARIES

- `Arduino_Modulino` — pulls in the sensor-vendor dependencies automatically
- `LiquidCrystal_I2C` — Library Manager

CODE REFERENCE

Range → bar graph

`constrain()` clips the reading first so `map()` can't produce an out-of-range count.

```
// --- Inside loop() ---
int clamped =
  constrain(mm, NEAR_MM, FAR_MM);
int lit =
  map(clamped, FAR_MM, NEAR_MM, 0, 8);
```

Beeping without `delay()`

Checked every loop, but only fires once `interval` ms have actually passed.

```
// --- Variables (top of sketch) ---
unsigned long lastBeepTime = 0;

// --- Inside loop() ---
if (millis() - lastBeepTime >= interval) {
  lastBeepTime = millis();
  buzzer.tone(freq, 50);
}
```

TRY IT YOURSELF

- Show "too close!" on the LCD once the distance drops below the solid-tone threshold.
- Fade the color smoothly from green to red instead of three fixed steps.